



UAlg

UNIVERSIDADE DO ALGARVE

Metaheuristics

MAXSAT Problem Solver

An Exhaustive Search Approach

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Questions 1, 2 and 3 are for learning the “cnf” structure and building the actual program.

4. Run your algorithm on the “hoos.cnf” instance and list all solutions that satisfy the problem instance.

Algorithm execution time: 0.000086 seconds

Best Result: 6

Best Hypothesis: ['11000', '11001']

5. Run your algorithm on the uf20-01.cnf, uf20-03.cnf, and uf20-03.cnf instances. How many solutions satisfy each problem instance?

“uf20-01.cnf”:

Algorithm execution time: 3.013765 seconds

Best Result: 91

Best Hypothesis: ['01110001111001101111', '10000100000011101001',
'10000100100001101001', '10000100100011101001', '10010000010011101001',
'10010001010011101001', '10010100000011101001', '10010100010011101001']

Number of solutions: 8

“uf20-02.cnf”:

Algorithm execution time: 2.468234 seconds

Best Result: 91

Best Hypothesis: ['00000011000001010010', '00000011000001110010',
'00000011100001010010', '00000011100001110010', '00001011100001010010',
'00001011100001110010', '00001011100101010010', '00001011100101110010',
'00001111100101110010', '00100011000001010010', '00100011000001110010',
'00100011100001010010', '00100011100001110010', '00101011100001010010',
'00101011100001110010', '00101011100101010010', '00101011100101110010',
'00101111100101110010', '10000011100001010000', '10000011100001010010',
'10000011100001110000', '10000011100001110010', '10001011100001010000',
'10001011100001010010', '10001011100001110000', '10001011100001110010',
'10001011100101010010', '10001011100101110010', '10001111100101110010']

Number of solutions: 29

“uf20-03.cnf”:

Algorithm execution time: 5.480428 seconds

Best Result: 91

Best Hypothesis: ['11110111111010011101']

Number of solutions: 1

6. It is impractical to do the same for a larger instance, say one contained in uf100-430. How could you estimate the time needed by your computer program to complete the task on such an instance? Justify your answer.

We can estimate the time needed for the computer to complete the task if we just use the “rule of three” and the right scale. We can use the number of possible hypotheses as the X and the time for completion as the Y. So:

Given that the average time for $2^{20} = 1\,048\,576$ hypotheses is 3,654142 seconds (collected from the 3 sets above) we can calculate by doing:

$$3,654142 * 2^{100} / 2^{20} = 4,42 \times 10^{27} \text{ seconds}$$

Converting the result, given or take:

- $5.12 * 10^{22}$ days
- $1.40 * 10^{20}$ years
- $1.40 * 10^{19}$ decades

Therefore, you can see that this exhaustive search is not applicable to instances of 100 variables.